MANUFACTURING / PROCESS DESCRIPTION

(For MPCB – Consent to Establish / Consent to Operate)

1. Introduction

The proposed unit is designed for processing of segregated organic / agricultural waste into value-added products such as biomass pellets/briquettes, & liquid organic fertilizer (LOF).

The plant is designed with a dry and semi-dry processing approach, minimising water consumption and operating under Zero Liquid Discharge (ZLD) conditions.

The proposed plant capacity is 5 TPD.

2. Raw Material

- Segregated organic waste
- Agricultural residues
- Biomass waste (fruits & vegetable waste, sawdust, agro residues, etc.)

All raw materials are received in segregated form and stored in a designated covered raw material storage area.

3. Detailed Manufacturing Process

(As per attached Process Flow Chart)

3.1 Segregation & Feeding

Segregated organic waste is manually inspected to remove any unwanted inert material such as stones, plastic, or metal pieces. The segregated waste is then fed into the processing line.

3.2 Shredding / Chopping

The segregated organic waste is subjected to shredding / chopping using a shredder. This step reduces the size of the waste material to ensure uniform processing and improved drying efficiency.

3.3 Mechanical Dewatering

The mixed organic material is passed through a mechanical dewatering unit to remove excess free moisture.

- Separated liquid (leachate) is collected in a closed system.
- The collected leachate is stabilized and converted into Liquid Organic Fertilizer (LOF).

No untreated wastewater is discharged outside the premises.

3.4 Mixing

Shredded material is transferred to a mixing unit, where uniform blending of organic waste is carried out. Mixing ensures consistency in moisture content and particle size, which is essential for downstream processes such as drying and densification.

3.5 Sun Drying / Thermal Drying

After mechanical dewatering, the solid biomass undergoes drying through:

- Sun drying, wherever feasible, and/or
- Thermal drying, using a controlled drying system

Thermal drying reduces moisture content to the required level for pelletization or briquetting. Exhaust gases from the dryer are passed through cyclone separator, water scrubber, and bag filter to control particulate emissions.

3.6 Biomass Grinding

Dried biomass is further processed through a Hammer Mill unit to achieve the desired particle size. Uniform particle size improves densification efficiency and product quality.

3.7 Pelletization / Briquetting

The processed biomass is fed into:

- Pellet machines for pellet production, and/or
- Briquette machines for briquette production

High pressure and mechanical compaction convert loose biomass into dense, uniform pellets or briquettes without the use of any chemical binders.

3.8 Product Storage & Dispatch

Finished products such as:

- Biomass pellets
- Biomass briquettes

are stored in a covered storage area on pucca flooring and dispatched as per market demand.

3.9. Liquid Organic Fertilizer (LOF) Generation

Leachate / condensate collected during mechanical dewatering and scrubbing is:

- Collected separately
- Stabilized and processed
- Converted into Liquid Organic Fertilizer (LOF)

This ensures resource recovery and eliminates liquid waste discharge.

4. Utilities

- Water: Used only for domestic purpose, floor cleaning, mixing in LOF and dust suppression
- Electricity: Used for operation of shredders, dryers, pellet/briquette machines, blowers, etc.
- Fuel: Biomass-based fuel -own biomass pellets

5. Environmental Safeguards

- No industrial effluent generation
- Zero Liquid Discharge (ZLD) system
- Adequate air pollution control devices
- Proper solid waste handling and reuse
- Noise and odour control measures implemented

6. Conclusion

The proposed manufacturing process is environment-friendly, resource-efficient, and compliant with MPCB norms. The unit ensures minimum pollution, effective waste utilization, and conversion of organic waste into useful products such as biomass fuel, and liquid organic fertilizer.

For Akshay Sansaadhan Pvt. Ltd.





Authorized Signatory
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Designation: Founder & CEO
Date: 01-02-2026
Place: Pune